VFX and audio

VFX artist, sound designer

`scenario-roblox-vfx-audio`

Builds layered Roblox effects that the server decides and each client draws, and wires Audio API graphs proven connected and measured, leaving only the listening to a person.



Live capture, Studio 0.743: the Boom effect frozen from 50 to 1,600 ms, empty by 1.2 s.



Layered slash effect being tuned in Studio. Twist VFX, How to Make Slash VFX, 10:57.

USE IT WHEN

- Making slash, impact, fireball or aura effects with particles, beams and trails
- Effects lag, pile up, stay invisible, or only one player sees them
- No sound, silent 3D audio, ducking, footsteps, or importing Scenario audio

WHAT THE EXPERTS INSIST ON

- If you can make out the impact flash's shape, it lasts too long: about 0.05 s. Twist VFX (Roblox VFX artist)
- Stack several texture layers and emit each once; bursts use Emit, only lingering tails use Enabled. MoldySugar (Roblox VFX artist)
- Impacts need the server and its raycast: an effect made locally is seen only by that player. Develuper (Roblox dev educator)
- A silent world is a missing listener: without an AudioListener or AudioDeviceOutput, nothing is heard. Cookie Tech (scripting educator)
- Footsteps follow animation events on contact frames, with a few material sets, not one looped sound. GnomeCode (scripting educator)

WORKFLOW, WITH THE GATE BEFORE EACH NEXT STAGE

1. **Set up the place.** **Gate:** kit modules present; the audio audit finds a listener
2. **Build templates as data.** **Gate:** VfxAudit reports 0 errors and 0 warnings
3. **Review frozen frames.** **Gate:** frames differ, the first frame reads, nothing left in the last
4. **Replicate through the server.** **Gate:** server holds 0 effect copies; client copies return to 0
5. **Wire the audio graph.** **Gate:** every Wire connected, path reaches the output, analyzer reads a level
6. **Mix ducking and footsteps.** **Gate:** reduction measured in dB; steps per distance and set per material
7. **Hand over the ear pass.** **Gate:** a listening list with place and timestamps goes to a person

KEY NUMBERS

Impact flash lifetime	**about 0.05 s**
Slash layer lifetime and speed	**0.2 to 0.3 s; Speed 0.001 to 0.01, never 0**
Beam semicircle CurveSize	**chord x 2/3; a 20-stud chord gives 13.33**
Particle emitter rate cap	**400 per s; 100 per s on mobile**
Ducking start on the music bus	**Threshold -30 dB, Ratio 8; -21 dB measured**
Audio file import limits	**under 20 MB, under 7 min, at most 48 kHz**

CHECKS THE AGENT RUNS ON ITS OWN WORK

- VfxAudit and AudioAudit clean; every Wire.Connected true; path reaches AudioDeviceOutput
- Analyzer RmsLevel above 0 while playing; GetAudibilityFor matches the intended curve
- Playtest: server holds no effect instances; client copies back to 0 after lifetime
- Freeze-frame contact sheets judged with the critique rubric at a raised Editor Quality Level

HANDOFFS

Receives VFX textures and flipbook frames from scenario-game-assets; sounds and music from scenario-audio; hit events from gameplay.

Delivers Effect templates, a client replicator, a wired audio graph, checked audio files, and an ear-pass list for a person.

TOOLKIT `RBX_VFX_AUDIO.PY`

- `timeline()`: freeze an effect at set times and capture each
- `make_flipbook()`: pack frames into a 1024 px grid sheet
- `prepare_audio()`: trim, resample to 48 kHz and level a file

MAIN EXPERT SOURCES

Twist VFX
Roblox VFX artist, self-declared non-scripter, How to Make Slash VFX (2026)

Develuper
Roblox dev educator, How to ACTUALLY Use VFX (2026)

Cookie Tech
Scripting educator, How to use the brand new audio API (2024)

GnomeCode
Scripting educator, Custom Footstep Sounds and Effects (2022)

STATUS

Tests 5 offline checks pass

In app 17 live tests pass in Roblox Studio 0.743 (two on a rerun after Studio timeouts)

G5 lighting, VFX and audio

without skills	9/15
with skills	14/15